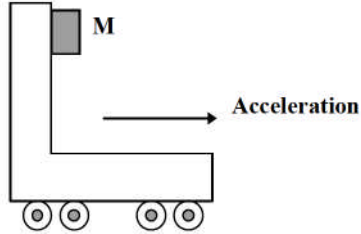
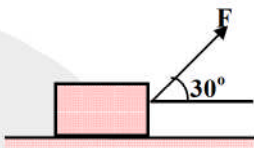
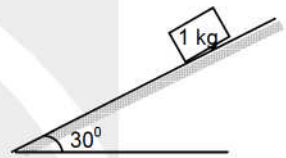
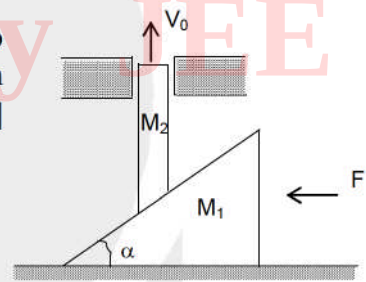
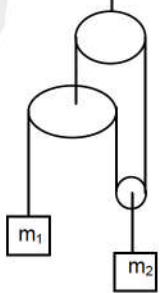
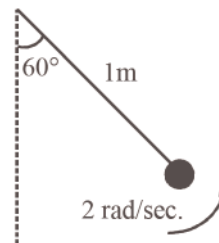


[SINGLE CORRECT CHOICE TYPE]

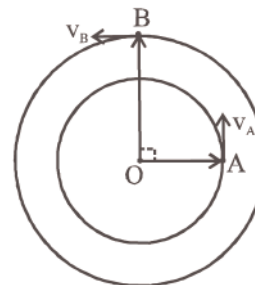
- Q.1 An accelerated system with a vertical wall has co-efficient of friction μ between block and walls as shown in the figure. A block M of mass 1 kg just remains in equilibrium with the vertical wall, when the system has an acceleration of 20 m/s^2 . The co-efficient of friction has a value [3]
- (A) 0.10 (B) 0.25 (C) 0.50 (D) 1
- 
- Q.2 A mass m rests on a horizontal surface. The coefficient of friction between the mass and the surface is μ . A force F is acting on the body as shown in the figure. The force of friction on mass m by surface is [3]
- (A) μmg (B) $F \frac{\sqrt{3}}{2}$ (C) $\mu [mg - \frac{\sqrt{3}}{2} F]$ (D) $\mu [mg + \frac{\sqrt{3}}{2} F]$
- 
- Q.3 A block of mass 1 kg is accelerating downwards on an rough inclined plane of inclination 30° with acceleration 2.5 m/s^2 . Find the contact force between the block and surface. ($g = 10 \text{ m/s}^2$) [3]
- (A) $10\sqrt{3} \text{ N}$ (B) $5\sqrt{3} \text{ N}$ (C) $\frac{5\sqrt{3}}{2}$ (D) none of these
- 
- Q.4 The force F is applied horizontally on the wedge of mass M_1 so as to raise the rod of mass M_2 upwards with a constant velocity v_0 as shown in the figure. The value of F is (neglect friction at the contact surfaces) [3]
- (A) $M_1 g \tan \alpha$ (B) $M_2 g \tan \alpha$
(C) $(M_1 + M_2) g \cos \alpha$ (D) $\left(\frac{M_1 M_2}{M_1 + M_2} \right) g \cos \alpha$
- 
- Q.5 A light inextensible string has a breaking strength of w newtons. With what maximum acceleration it can pull a block of weight $\frac{w}{5}$ newtons vertically upwards. [3]
- (A) $4g$ (B) $5g$ (C) $g/5$ (D) $4g/5$
- 
- Q.6 Pulley mass system is shown in figure pulleys and strings are ideal. acceleration of m_2 and m_1 [3]
- (A) $a_1 = 2a_2$ (B) $a_2 = 2a_1$
(C) $a_1 = a_2 = g$ (D) data insufficient
- 
- Q.7 Two men P & Q of masses M and $m (< M)$ hold a light rope passing over a smooth fixed pulley P and Q climb up the rope so that the acceleration of Q upward is double that of P downward. The tension in the rope is [3]
- (A) $\frac{2Mm}{M+m} g$ (B) $\frac{3Mm}{M+2m} g$ (C) $\frac{3Mm}{2(M+m)} g$ (D) $\frac{3Mm}{2M+m} g$

- Q.8 A simple pendulum of mass 0.5 kg, during its swing in the vertical plane, is observed to have an angular velocity of 2 rad/s in the position shown. The instantaneous tension in the string is ($g = 10 \text{ m/s}^2$) [3]
 (A) 2.5 N (B) 4 N
 (C) 4.5 N (D) 5 N



- Q.9 Two particles A and B are moving on two different concentric circles of radii r and $\sqrt{3}r$ with different velocities $v_A = v$ and $v_B = 2v$ respectively. Angular velocity of B relative to A as observed by A is: [3]

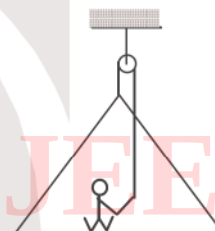
(A) $\frac{(2\sqrt{3}+1)v}{4r}$ (B) $\frac{v}{2r}$ (C) $\frac{(2-\sqrt{3})v}{\sqrt{3}r}$ (D) $\frac{(2\sqrt{3}-1)v}{r}$



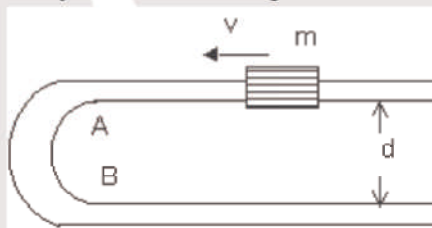
- Q.10 A particle of mass 'm' originally at rest, is subjected to a force whose direction is constant but whose magnitude varies with according to the relation $F = F_0 t \left[\frac{2}{T} - \frac{t}{T^2} \right]$ where F_0 and T are constants. Then speed of the particle after a time $2T$ is [3]

(A) $\frac{3F_0 T}{4m}$ (B) $\frac{4F_0 T}{3m}$ (C) $\frac{2F_0 T}{5m}$ (D) $\frac{F_0 T}{m}$

- Q.11 A man of mass 50 kg stands on a frame of mass 30 kg he pulls on a light rope which passes over a pulley. The other end of the rope is attached to the frame. For the system to be in equilibrium what force man must exert on the rope [3]
 (A) 40 g (B) 80 g
 (C) 30 g (D) 50 g

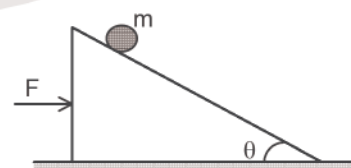


- Q.12 A U shaped smooth wire has a semi-circular bending between A and B as shown in the figure. A bead of mass 'm' moving with uniform speed v through the wire enters the semicircular bend at A and leaves at B. The average force exerted by the bead on the part AB of the wire is, [3]



(A) 0 (B) $\frac{4mv^2}{\pi d}$ (C) $\frac{2mv^2}{\pi d}$ (D) none of these.

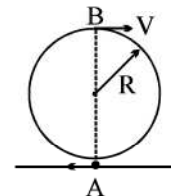
- Q.13 In the given figure, the wedge is acted on by a constant horizontal force F . The wedge is moving on a smooth horizontal surface. A ball of mass 'm' is at rest relative to the wedge. The ratio of forces exerted on 'm' by the wedge when F is acting and force exerted by wedge when it is at rest. Assuming no friction between the wedge and the ball, is equal to [3]
 (A) $\sec^2 \theta$ (B) $\cos^2 \theta$ (C) 1 (D) none of these.



- Q.14 A body is thrown with a velocity of 10 m/s at an angle of 45° to the horizontal. The radius of curvature of its trajectory in $t = 1/\sqrt{2}$ sec after the body began to move is : [3]
 (A) 0 m (B) 2.5 m (C) 5 m (D) None

- Q.15 Particle A is moving in a horizontal plane with constant velocity V as shown. Another particle B is moving in a circle with same speed V . At the moment when A is diametrically opposite to B, the radius of curvature of B as seen by A will be (Radius of circle is R) [3]

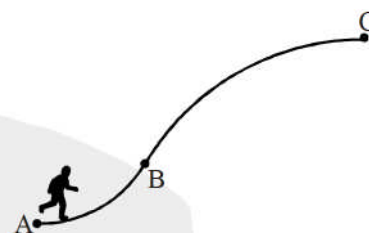
(A) R (B) $R/2$
(C) $2R$ (D) $4R$



[PARAGRAPH TYPE]

Paragraph for question nos. 16 to 18

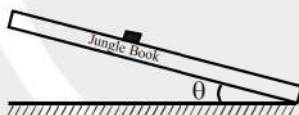
An athlete of mass 80 kg is running on a rough track whose cross-section is shown below. The lower part AB of track is a cylindrical valley of radius 100 m and upper part BC is a cylindrical hill of radius 200 m . The two parts join such that there is no sudden change of slope of the track. The speed of the athlete on the track is always 5 m/s . A is the lowest point of the valley, B is the point at which valley ends and hill starts and C is the top of the hill. While moving from A to C, the athlete travels a horizontal distance of 150 m .



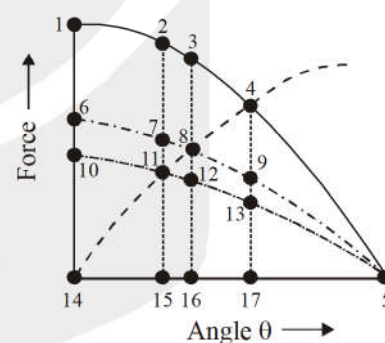
- Q.16 Find the time taken by the athlete in going from A to C [3]
(A) $5\pi \text{ sec}$ (B) $10\pi \text{ sec}$ (C) $15\pi \text{ sec}$ (D) $20\pi \text{ sec}$
- Q.17 The correct order of normal force experienced by athlete is : [3]
(A) $N_A > N_B > N_C$ (B) $N_A < N_B < N_C$ (C) $N_A > N_C > N_B$ (D) $N_C > N_A > N_B$
- Q.18 The magnitude of friction force experienced by the athlete is : [3]
(A) zero throughout (B) decreases continuously during motion from A to C
(C) increases continuously from A to C (D) attains a maximum value at B

[MULTIPLE CORRECT CHOICE TYPE]

Paragraph for question nos. 19 to 21



Given graph is a plot of magnitude of normal, limiting friction, kinetic friction and static friction versus angle of inclination θ for the coin lying on book. 17 points are labeled with numbers. The coin slips at a certain angle θ_c . The coefficient of static friction is 0.6 , and the coefficient of kinetic friction is 0.4 . First consider the range of θ for which the coin is stationary. Which points of the curves can represent



- Q.19 The magnitude of the normal force on the coin from the book, [4]
(A) 6 (B) 2 (C) 3 (D) 11
- Q.20 The magnitude of the static frictional force on the coin from the book [4]
(A) 8 (B) 4 (C) 11 (D) 9
- Q.21 Next, consider the range of θ for which the coin slides. Which points can give the magnitude of the kinetic frictional force on the coin from the book? [4]
(A) 12 (B) 13 (C) 11 (D) 8

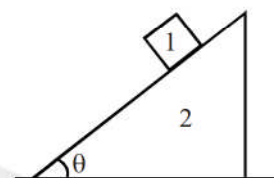
[MULTIPLE CORRECT CHOICE TYPE]

- Q.22 A heavy particle is tied to the end A of a string of length 1.6 m. Its other end O is fixed. It revolves as a conical pendulum with the string making 60° with the vertical. Then [4]

(A) its period of revolution is $\frac{4\pi}{7}$ sec.
 (B) the tension in the string is doubled the weight of the particle
 (C) the velocity of the particle = $2.8\sqrt{3}$ m/s
 (D) the centripetal acceleration of the particle is $9.8\sqrt{3}$ m/s².

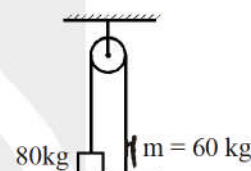
- Q.23 Block-1 and wedge 2 can move relative to each other. Block-1 has mass m and wedge-2 has also some mass. Inclined plane inclination is θ . There is friction coefficient μ between 1 and 2. Ground is smooth. 1 and 2 are released from rest to move then choose the correct statement(s). [4]

(A) If $\mu > \tan \theta$ then block-1 and wedge-2 both will not move.
 (B) If $\mu > \tan \theta$ then block-1 will not move w.r.t. wedge-2 but wedge-2 will move.
 (C) If $\mu < \tan \theta$ then block-1 and wedge-2 both will move.
 (D) If $\mu < \tan \theta$ then block-1 will move w.r.t. wedge-2 but wedge-2 will not move.



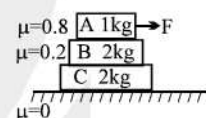
- Q.24 A man of mass 60 kg starts climbing up a massless rope by exerting 900 N on the string. After 2 sec, he is exhausted. He stops climbing and holds on to the rope. [4]

(A) String will become slack for some time.
 (B) The speed of man at end of 2 sec. is 10 m/s
 (C) Just after $t = 2$ sec, the acceleration of man & block will be same in magnitude
 (D) The acceleration of man just after 2 sec is $g/7$.



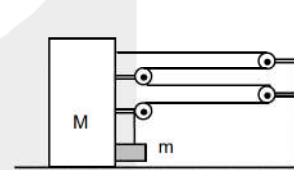
- Q.25 Three blocks lie on each other as shown. Horizontal force F (variable) is applied on block A. Choose the correct statements. ($g = 10$ m/s²) [4]

(A) maximum acceleration of block B can be 2 m/s².
 (B) slipping between B and C cannot take place.
 (C) slipping between A and B starts when F is more than 8 N
 (D) maximum acceleration of block C can be 3 m/s².



[SUBJECTIVE TYPE]

- Q.26 A block of mass M is connected with a particle of mass m by a light inextensible string as shown in figure. Assuming all contacting surfaces as smooth, find the acceleration of the wedge after releasing the system. [5]



- Q.27 A car is moving in a circular path of radius 50 m, on a flat rough horizontal ground. The mass of the car is 1000 kg. At a certain moment, when the speed of the car is 5 m/s, the driver is increasing speed at the rate of 1 m/s². Find the value of static friction on tyres (total) at this moment, in Newtons. [5]

ANSWER KEY

- | | | | | | | | | | | | | | |
|------|-------------------------|------|---------------|------|-------|------|-----|------|-----|------|-----|------|-----|
| Q.1 | C | Q.2 | AB | Q.3 | D | Q.4 | B | Q.5 | A | Q.6 | C | Q.7 | B |
| Q.8 | C | Q.9 | A | Q.10 | B | Q.11 | A | Q.12 | B | Q.13 | A | Q.14 | C |
| Q.15 | D | Q.16 | B | Q.17 | C | Q.18 | D | Q.19 | B,C | Q.20 | A,C | Q.21 | A,B |
| Q.22 | A,B,C,D | Q.23 | A, C | Q.24 | A,B,C | Q.25 | A,B | | | | | | |
| Q.26 | $a = \frac{4mg}{M+17m}$ | Q.27 | $500\sqrt{5}$ | | | | | | | | | | |